

3.7. Advanced Multi-Step Form Using Oracle JET Components

Introduction

This practice demonstrates how to build a rich, user-friendly **multi-step form** using Oracle JET. It incorporates key UI components like `oj-train`, `oj-form-layout`, `oj-input-text`, `oj-select-single`, `oj-avatar`, and `oj-dialog`. The form follows a progressive step-by-step flow that captures personal information, user preferences, and then displays a summary for review and submission.

Objective

By the end of this practice, learners will be able to:

- Build a dynamic multi-step form using Oracle JET's **oj-train** component.
- Capture and validate user input using **form layout**, **input**, **radio**, **checkbox**, and **select** components.
- Create **conditional dropdowns** (Country → State) using `ko.computed` and `ArrayDataProvider`.
- Implement **avatar initials** logic using observables.
- Use **file picker** (`oj-c-file-picker`) to select and display a profile image name.
- Handle form **validation**, **navigation**, **reset**, and **submission** using `oj-dialog`.
- Bind and show submitted data as **JSON** using knockout.

Prerequisites

To complete this practice, learners should have:

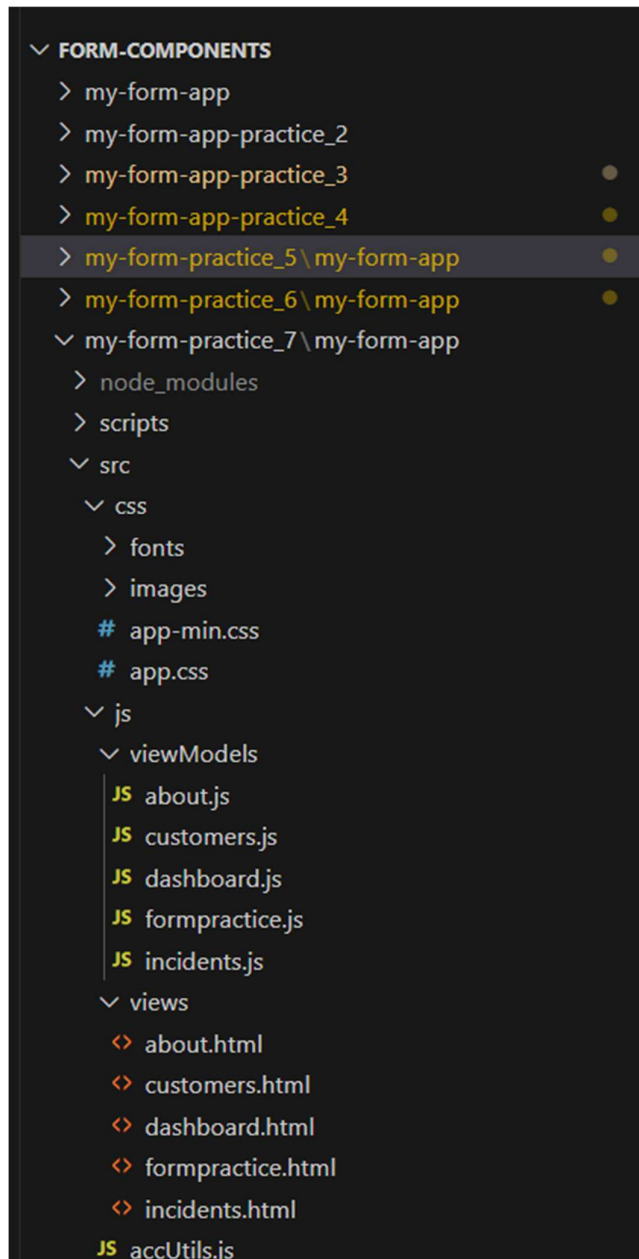
- Basic understanding of **Oracle JET** framework and component lifecycle.
- Familiarity with **KnockoutJS** (`ko.observable`, `ko.computed`).
- Experience with **MVVM architecture** in Oracle JET applications.
- Working knowledge of **oj-module** routing structure (e.g., `formpractice.js` and `formpractice.html`).

Ensure all the following are complete:

- Practice 1: Name and Email fields
- Practice 2: Country dropdown with `oj-select-single`
- Practice 3: Gender (`oj-radioset`) and Hobbies (`oj-checkboxset`)
- Practice 4: creating submit button

- Practice 5 : creating train, avatars etc.
- You are working in a JET app (e.g., NavDrawer template)

Project Structure:



Tasks:

1. Create a new folder named " my-form-app-practice_7" and move to that directory
2. In the command line run, `Ojet create my-form-app --template=navdrawer`

3. Then run **ojet build web** and **ojet serve** respectively (like we did before)
4. Create folders `/js/viewModels/formpractice.js` and `/js/views/formpractice.html`
5. Add code to `formpractice.js`

```
JS formpractice.js X
my-form-practice_7 > my-form-app > src > js > viewModels > JS formpractice.js > define() callback > Fo
1  define([
2      'knockout',
3      'ojs/ojarraydataprovider',
4      'ojs/ojformlayout',
5      'ojs/ojinputtext',
6      'ojs/ojselectsingle',
7      'ojs/ojradioset',
8      'ojs/ojcheckboxset',
9      'ojs/ojbutton',
10     'ojs/ojdialog',
11     'ojs/ojmessages',
12     'ojs/ojtrain',
13     'ojs/ojavatar',
14     'ojs/ojlabel',
15     'oj-c/file-picker'
16 ], function (ko, ArrayDataProvider) {
17     function FormPracticeViewModel() {
18         this.trainSteps = [
19             { id: 'step1', label: 'Personal Information' },
20             { id: 'step2', label: 'User Preferences' },
21             { id: 'step3', label: 'Review and Submit' }
22         ];
23         this.stepValue = ko.observable('step1');
24
25         // Step 1: Personal Info
26         this.name = ko.observable('');
27         this.email = ko.observable('');
28         this.selectedCountry = ko.observable('');
29         this.selectedState = ko.observable('');
30
31         // Step 2: Preferences
32         this.gender = ko.observable('');
33         this.hobbies = ko.observableArray([]);
34         this.profilePhoto = ko.observableArray([]);
35
36         this.acceptArr = ko.observableArray([]);
37         this.multipleStr = ko.observable('single');
```

```
38     this.isDisabled = ko.observable(false);
39
40     this.countryOptions = [
41       { value: 'IN', label: 'India' },
42       { value: 'US', label: 'USA' },
43       { value: 'UK', label: 'UK' }
44     ];
45     this.countryDataProvider = new ArrayDataProvider(this.countryOptions, { keyAttributes: 'value' });
46
47     this.stateMap = {
48       IN: [
49         { value: 'PB', label: 'Punjab' },
50         { value: 'DL', label: 'Delhi' },
51         { value: 'MH', label: 'Maharashtra' }
52       ],
53       US: [
54         { value: 'NY', label: 'New York' },
55         { value: 'CA', label: 'California' },
56         { value: 'TX', label: 'Texas' }
57       ],
58       UK: [
59         { value: 'ENG', label: 'England' },
60         { value: 'SCT', label: 'Scotland' },
61         { value: 'WLS', label: 'Wales' }
62       ]
63     };
64
65     this.stateDataProvider = ko.pureComputed(() => {
66       const country = this.selectedCountry();
67       const states = this.stateMap[country] || [];
68       return new ArrayDataProvider(states, { keyAttributes: 'value' });
69     });
70
71     this.isStateEnabled = ko.computed(() => !!this.selectedCountry());
```

```

72
73     this.avatarInitials = ko.computed(() => {
74         const name = this.name();
75         if (!name) return '';
76         const parts = name.trim().split(' ');
77         if (parts.length === 1) return parts[0].charAt(0).toUpperCase();
78         if (parts.length >= 2) return (parts[0].charAt(0) + parts[parts.length - 1].charAt(0)).toUpperCase();
79         return '';
80     });
81
82     this.formSubmitted = ko.observable(false);
83     this.submittedData = ko.observable({});
84
85     this.goToNextStep = () => {
86         if (this.stepValue() === 'step1') {
87             const form1 = document.getElementById('personalInfoForm');
88             if (form1 && !form1.checkValidity()) {
89                 form1.reportValidity();
90                 return;
91             }
92             this.stepValue('step2');
93         } else if (this.stepValue() === 'step2') {
94             this.stepValue('step3');
95         }
96     };
97
98     this.goToPreviousStep = () => {
99         if (this.stepValue() === 'step2') {
100             this.stepValue('step1');
101         } else if (this.stepValue() === 'step3') {
102             this.stepValue('step2');
103         }
104     };
105

```

```

106     this.selectListener = (event) => {
107         console.log("Files selected:", event.detail.files);
108         this.profilePhoto(event.detail.files);
109     };
110
111     this.invalidListener = (event) => {
112         console.log("Invalid file selected:", event.detail);
113     };
114
115     this.handleSubmit = (event) => {
116         if (event) {
117             event.preventDefault();
118         }
119
120         if (this.stepValue() === 'step3') {
121             const formData = {
122                 name: this.name(),
123                 email: this.email(),
124                 country: this.selectedCountry(),
125                 state: this.selectedState(),
126                 gender: this.gender(),
127                 hobbies: this.hobbies(),
128                 profilePhotoName: this.profilePhoto().length > 0 ? this.profilePhoto()[0].name : 'None selected'
129             };
130
131             console.log("Form Submitted:", formData);
132             this.submittedData(formData);
133             this.formSubmitted(true);
134
135             const dialog = document.getElementById('submitDialog');
136             if (dialog) {
137                 dialog.open();

```

```

137         dialog.open();
138     }
139 }
140 };
141
142 this.resetForm = () => {
143     this.name('');
144     this.email('');
145     this.selectedCountry('');
146     this.selectedState('');
147     this.gender('');
148     this.hobbies([]);
149     this.profilePhoto([]);
150     this.formSubmitted(false);
151     this.submittedData({});
152     this.stepValue('step1');
153 };
154
155 this.closeDialog = () => {
156     const dialog = document.getElementById('submitDialog');
157     if (dialog) {
158         dialog.close();
159     }
160 };
161
162 this.handleTrainStepChange = (event) => {
163     this.stepValue(event.detail.value);
164 };
165 }
166
167 return new FormPracticeViewModel();
168 });

```

6. now add code for formpractice.html

```

my-form-practice_7 > my-form-app > src > js > views > formpractice.html > div.oj-hybrid-padding > style
1  <div class="oj-hybrid-padding">
2  <style>
3      .avatar-container {
4          position: absolute;
5          top: 130px;
6          right: 100px;
7      }
8      .step-content {
9          margin-top: 20px;
10     }
11 </style>
12
13 <div class="oj-flex">
14     <oj-train id="profileTrain"
15         steps="[[trainSteps]]"
16         selected-step="{{stepValue}}"
17         on-selected-step-changed="[[handleTrainStepChange]]">
18         <oj-train-step id="step1" label="Personal Information"></oj-train-step>
19         <oj-train-step id="step2" label="User Preferences"></oj-train-step>
20         <oj-train-step id="step3" label="Review and Submit"></oj-train-step>
21     </oj-train>
22 </div>
23
24 <div class="step-content" data-bind="if: stepValue() === 'step1'">
25     <h2>Enter Your Personal Details</h2>
26     <form id="personalInfoForm" class="oj-form-layout" max-columns="2" direction="row" onsubmit="return false;">
27         <oj-input-text label-hint="Enter name" required="true" value="{{(name)}}"></oj-input-text>
28         <oj-input-text label-hint="Enter email" required="true" value="{{(email)}}"></oj-input-text>
29
30         <oj-select-single label-hint="Select Country"
31             data="[[countryDataProvider]]"
32             value="{{selectedCountry}}"
33             required="true">
34         </oj-select-single>
35
36         <oj-select-single label-hint="Select State"
37             data="[[stateDataProvider]]"

```

```

38             value="{{selectedState}}"
39             required="true"
40             disabled="[[!isStateEnabled()]]">
41         </oj-select-single>
42     </form>
43 </div>
44
45 <div class="step-content" data-bind="if: stepValue() === 'step2'">
46     <h2>Tell Us About Your Preferences</h2>
47     <oj-form-layout>
48     <div>
49         <oj-label for="genderGroup">Gender</oj-label>
50         <oj-radioset id="genderGroup" value="{{gender}}" required="true">
51             <oj-option value="male">Male</oj-option>
52             <oj-option value="female">Female</oj-option>
53             <oj-option value="other">Other</oj-option>
54         </oj-radioset>
55     </div>
56
57     <div>
58         <oj-label for="hobbiesGroup">Hobbies</oj-label>
59         <oj-checkboxset id="hobbiesGroup" value="{{hobbies}}">
60             <oj-option value="reading">Reading</oj-option>
61             <oj-option value="sports">Sports</oj-option>
62             <oj-option value="music">Music</oj-option>
63         </oj-checkboxset>
64     </div>
65
66     <div>
67         <oj-label for="profilePhoto">Profile Photo</oj-label>
68         <oj-c-file-picker

```

```

68     <oj-c-file-picker
69         accept="[[acceptArr]]"
70         selection-mode="[[multipleStr]]"
71         on-oj-select="[[selectListener]]"
72         on-oj-invalid-select="[[invalidListener]]"
73         disabled="[[isDisabled]]"
74         label="Upload"></oj-c-file-picker>
75     <span data-bind="text: profilePhoto().length > 0 ? profilePhoto()[0].name : ''"></span>
76 </div>
77 </oj-form-layout>
78 </div>
79
80 <div class="step-content" data-bind="if: stepValue() === 'step3'"
81 <h2>Review Your Information</h2>
82 <p><strong>Name:</strong> <span data-bind="text: name()"></span></p>
83 <p><strong>Email:</strong> <span data-bind="text: email()"></span></p>
84 <p><strong>Country:</strong> <span data-bind="text: selectedCountry()"></span></p>
85 <p><strong>State:</strong> <span data-bind="text: selectedState()"></span></p>
86 <p><strong>Gender:</strong> <span data-bind="text: gender()"></span></p>
87 <p><strong>Hobbies:</strong> <span data-bind="text: hobbies().join(', ')"></span></p>
88 <p><strong>Profile Photo:</strong> <span data-bind="text: profilePhoto().length > 0 ? profilePhoto()[0].name : 'No
89
90 </div>
91
92 <div class="avatar-container">
93     <oj-avatar id="userAvatar"
94         role="img"
95         aria-label="User avatar"
96         size="lg"
97         initials="[[avatarInitials]]">
98
99 </div>
100
101 <div class="oj-flex oj-sm-justify-content-center oj-sm-margin-top-4">
102     <oj-button data-bind="click: goToPreviousStep, visible: stepValue() !== 'step1'">Back</oj-button>
103     <oj-button data-bind="click: goToNextStep, visible: stepValue() !== 'step3'" class="oj-sm-margin-start-8">Next</oj-button>
104     <oj-button data-bind="click: goToNextStep, visible: stepValue() !== 'step3'" on-oj-action="[[resetForm]]" class="oj-sm-margin-start-8">Reset</oj-button>
105     <oj-button data-bind="click: goToNextStep, visible: stepValue() == 'step3'" on-oj-action="[[handlesubmit]]" >Submit Form</oj-button>
106 </div>
107
108 <oj-dialog id="submitDialog" dialog-title="Submission Successful" modality="modeless">
109     <div slot="body">
110         <p>Your form was submitted successfully!</p>
111     </div>
112     <div slot="footer">
113         <oj-button on-oj-action="[[closeDialog]]">OK</oj-button>
114     </div>
115 </oj-dialog>
116
117 <div class="oj-sm-margin-top-4" data-bind="if: formSubmitted">
118     <h3>Submitted Data (JSON)</h3>
119     <pre data-bind="text: JSON.stringify(submittedData(), null, 2)"></pre>
120 </div>
121 </div>

```

7. Run ojet serve on command terminal

7. Navigate to the Form Practice page in the browser.
8. Enter Name, Email, and select a Country, hobbies and gender etc. fill all the fields. you will notice the advanced features in the UI like avatar with initials, step by step form filling (with oj-train>,

ORACLE App Name john.hancock@oracle.com

Dashboard Incidents Customers About Form Practice

1 Personal Information 2 User Preferences 3 Review and Submit

Enter Your Personal Details

Enter name
richa jain

Enter email
richa@gmail.com

Select Country
India

Select State

Required

Next Reset

9. Click on next . here you will submit your preferences and hobbies via radio buttons and checkboxes. Also there is file picker , where you can select the file from your local system and upload it. At the bottom, there is reset, back and next button options.

1 Personal Information 2 User Preferences 3 Review and Submit

Tell Us About Your Preferences

Gender

☐ Male

☒ Female

☐ Other

Hobbies

☒ Reading

☐ Sports

☐ Music

Profile Photo

Drag and Drop
Select a file or drop one here

Back Next Reset

10. Click next. Here you can review your entered info in read only format. also. You have a submit form and back button options . from where you can go to the previous page and submit the form respectively. Also, notice the train options at the top, you are at 3rd page.



Review Your Information

Name: richa jain
Email: richa@gmail.com
Country: IN
State:
Gender: female
Hobbies: reading
Profile Photo: None selected

Submission Successful
Your form was submitted successfully!

OK

[Back](#) [Submit Form](#)

Submitted Data (JSON)

```
{
  "name": "richa jain",
  "email": "richa@gmail.com",
  "country": "IN",
  "state": "",
  "gender": "female",
  "hobbies": [
    "reading"
  ],
  "profilePhotoName": "None_selected"
}
```